

Microalgae Research Report

ABSTRACT:

Creating sustainable energy options and access to clean water are two major problems that need solutions. Microalgae research has been on the rise recently as a 3rd generation fuel source. One large pollutant in water sources like streams that causes eutrophication is agricultural runoff which tends to be high in nitrate and phosphate which are necessary for algae to grow. In order to address this, two questions are brought up, what strains are optimal for water treatment, and which ones produce the most lipids or biomass for fuel. To test this, two strains of algae were cultivated, nitrate and phosphate levels were measured throughout the growing process and dry weights were taken. It was found that between the two strains chosen *Desmodesmus Abundans* grew faster and denser than *Chlorella Vulgaris* and the growth of the algae is heavily influenced by the nitrate and phosphate concentrations

Introduction:

We live in a world where energy is precious. Every single part of our lives is touched in one way, shape, or form of energy and where we get this energy matters. It has become increasingly apparent that the methods that we use are either not enough or harmful to the environment. Even in cases that we generally think of as being more environmentally friendly such as 2nd gen fuel sources like ones based on corn or other crops can lead to

negative outcomes such as increased food prices (Hossain et al.). What this tells us is that we need to find better alternatives. One alternative to this would be third generation fuel sources such as algae. These have the potential to address the food water fuel nexus with an interconnected approach to both the water and fuel side and does not encroach as directly on food resources as much as ethanol from corn or sugarcane does.

There is also the issue of water contamination from agricultural runoff. This contamination mainly presents itself as elevated nitrogen and phosphorus levels in the water. These elevated nutrient levels can be particularly harmful as they can lead to eutrophication. The beneficial side to this however is that certain aquatic organisms such as algae use this as a nutrient source. This means that if we introduce algae to feed on the N and P contamination, we can effectively reduce the N and P levels while also producing a source of biofuel that wouldn't otherwise be a food source for people and therefore shouldn't affect food market prices.

In this article I will present findings on the efficiency of two algal strains at filtering out the two main contaminants of agricultural runoff: nitrate and phosphate. I will also be presenting findings on the difference in the lipid concentration for the two strains when exposed to different phosphate and nitrate levels.

Background/Literature Review:

Our world has increased demand for energy, and we are increasingly turning to fossil fuels as a source. Unfortunately, fossil fuels also happen to release carbon dioxide into the

air. This release of greenhouse gas, CO₂, leads to an increase in atmospheric heat capacity which also means an increase in global warming.

Another big challenge the world faces is having clean water sources. Our planet is covered in water, up to 70% yet only 3% is drinkable (Rana et al.). Many fertilizers have nutrients that can leach into nearby water sources and wreak havoc on both the local ecosystems and on the inhabitants that drink the water of that area.

Microalgae have many beneficial characteristics for both energy production in the form of 3rd generation biofuel and water filtration. It can be grown in large bodies of water rather than on fields which would have them competing with crops for space and resources. Microalgae also depends mainly on C, P, N, and S for their essential nutrients, and, if grown phototrophically, they also utilize CO₂ as their carbon source (Zeng et al.). High levels of these nutrients are detrimental to water quality and are often the result of agricultural runoff from nitrogen-based fertilizers and farm animal manure. Excessive quantities of phosphate and nitrate can cause a process called eutrophication which leads to depletion of oxygen levels in water creating hypoxic dead zones.

Food, fuel, and water are all very interconnected resources in that you cannot solve an issue in one area of this nexus without it affecting at least one of the other two. For example, water is essential for the growth of food, food crops like corn can be used for fuel, and fuel is used in the harvesting of crops and the circulation of water through cities. If you are over farming and using larger amounts of water this can deplete water reservoirs much quicker and cause issues with crops next year or even cause issues in energy production

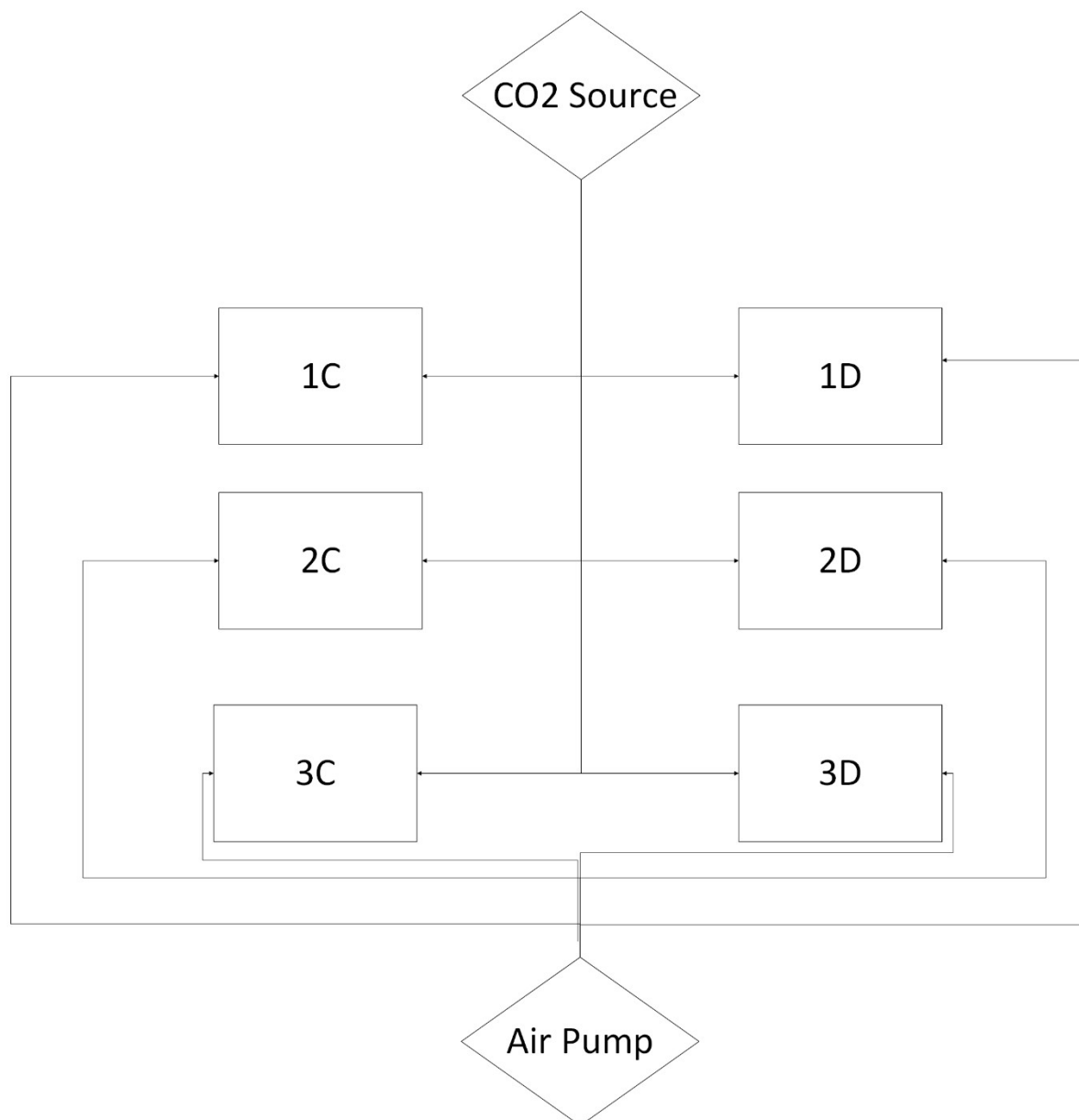
processes which use steam and cooling water. To address this interconnected issue microalgae functions to draw the nitrate and phosphate that has seeped in due to agricultural runoff out of the wastewater while also providing a lipid rich fuel source after cultivation.

Methods and Discussion/Analysis:

Microalgae was cultivated in 6 different 1L jars filled to 500mL. The algae was agitated by a aquarium grade air pump, and was fed CO₂ from a pressurized cannister. A control group was formed by using standard MLA media (Appendix B), a high phosphate group was made by adding 100% the original amount of Phosphate and the same for the high nitrate group. Sample groups were labeled with a C for *Chlorella Vulgaris* or D for *Desmodesmus Abundans*, 1 for control, 2 for high phosphate, and 3 for high nitrate. The setup is described below in Figure 1.

Figure 1.

Algae cultivation set up



UV spectrophotometry:

The UV spectrometer was used to measure the nitrate and phosphate levels of the liquid samples to determine by difference, the quantity of each contaminant that was removed by the algae as nutrient.

Centrifuge:

The centrifuge was used as a means for dewatering each of the samples so that it would be easier to extract the lipids.

Dry weight:

The dry weight was found so that there would be a way to determine which sample saw the most growth. Results are below in *Table 1*.

Table 1*Dry weight per volume*

Sample Number	Vol taken	Filter paper wt.	Dry wt.	Algae Dry weight (Dry wt. -Filter paper wt)	Dry wt. per Vol
1D 1	10 mL	0.1027 g	0.1139 g	0.0112 g	1.12 g/L
1D 2	10 mL	0.1045 g	0.1149 g	0.0104 g	1.04 g/L
2D 1	5 mL	0.1043 g	0.1145 g	0.0102 g	2.04 g/L
2D 2	5 mL	0.1030 g	0.1143 g	0.0113 g	2.26 g/L
3D 1	20 mL	0.1041 g	0.1157 g	0.0116 g	0.58 g/L
3D 2	20 mL	0.1029 g	0.1247 g	0.0218 g	1.09 g/L
1C 1	30 mL	0.1040 g	0.1443 g	0.0403 g	1.34 g/L
1C 2	20 mL	0.1028 g	0.1266 g	0.0238 g	1.19 g/L
2C 1	20 mL	0.1011 g	0.1170 g	0.0159 g	0.795 g/L
2C 2	20 mL	0.1023 g	0.1190 g	0.0167 g	0.835 g/L
3C 1	20 mL	0.1053 g	0.1146 g	0.0093 g	0.465 g/L
3C 2	20 mL	0.1038 g	0.1102 g	0.0064 g	0.32 g/L

Results and Discussion/Analysis

In *Figure 2 Comparison of dry weights per volume* and in Table 1, we can see that the sample that saw the most growth was 2D, which is the high phosphate *Desmodesmus Abundans* sample. This implies that in wastewater with high contamination levels of phosphate *Desmodesmus Abundans* would grow more densely and likely perform better. However, the one that performed the worst was the *Chlorella Vulgaris* in high nitrate conditions. This likely indicates that wastewater with a higher nitrate concentration would not be ideal media for cultivating *Chlorella Vulgaris* as a fuel source.

One way to get a better understanding of how the algae would perform as a fuel source in these conditions would be to run it again and extract and measure lipid concentrations of the different samples. Having more samples for redundancy and replicability would also be ideal. The nitrate and phosphate concentration readings from *Figures 3-14* do not reflect the expected downward trend expected of a nutrient source that the algae feeds off. This may indicate an error in either the method or the reading and may mean that in future work a different method should be used to measure phosphate and nitrate levels for more accuracy.

Figure 2

Comparison of dry weights per volume

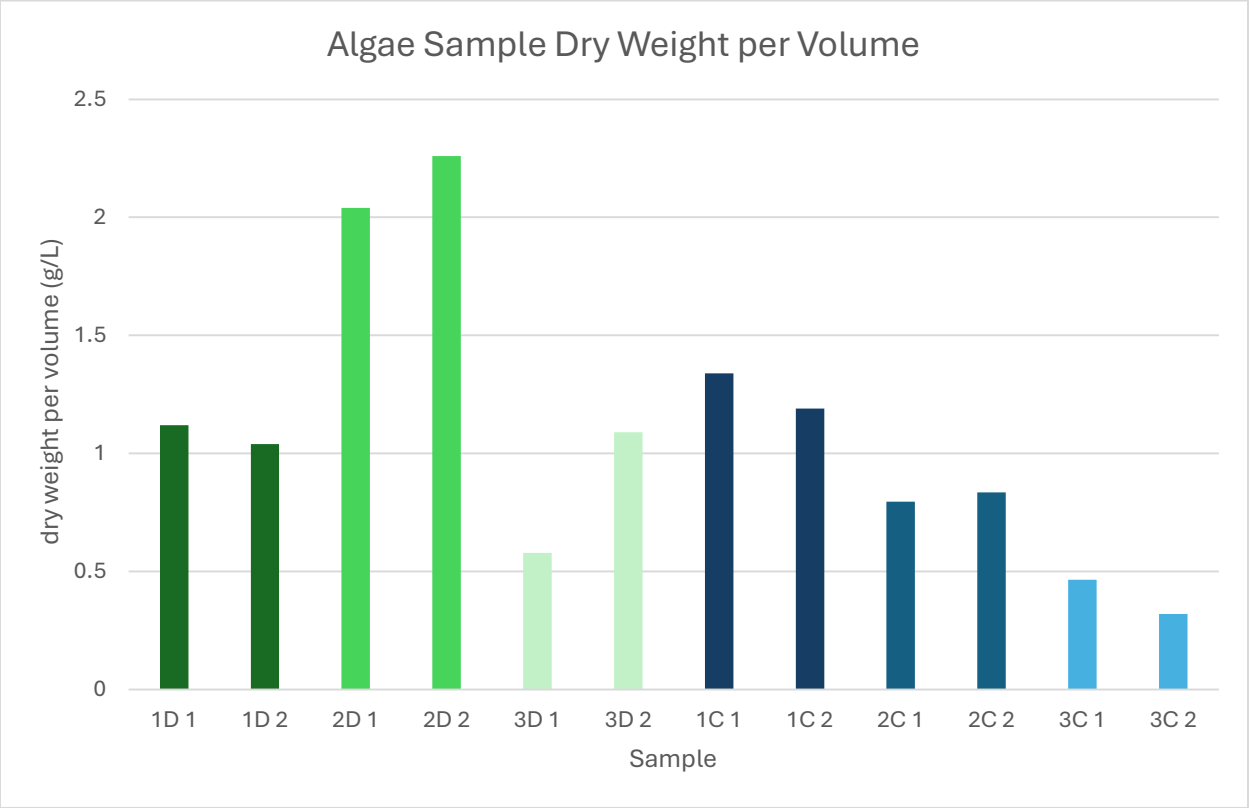


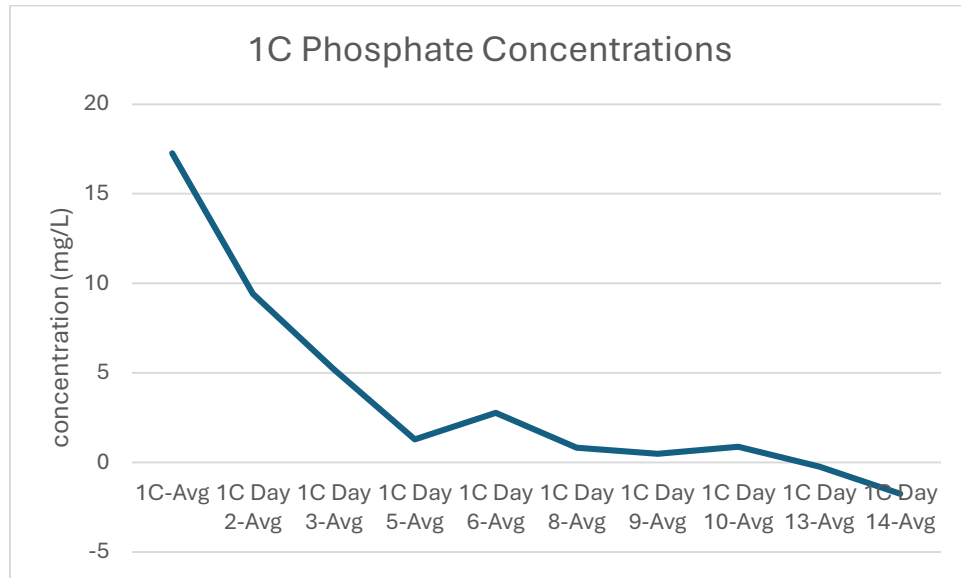
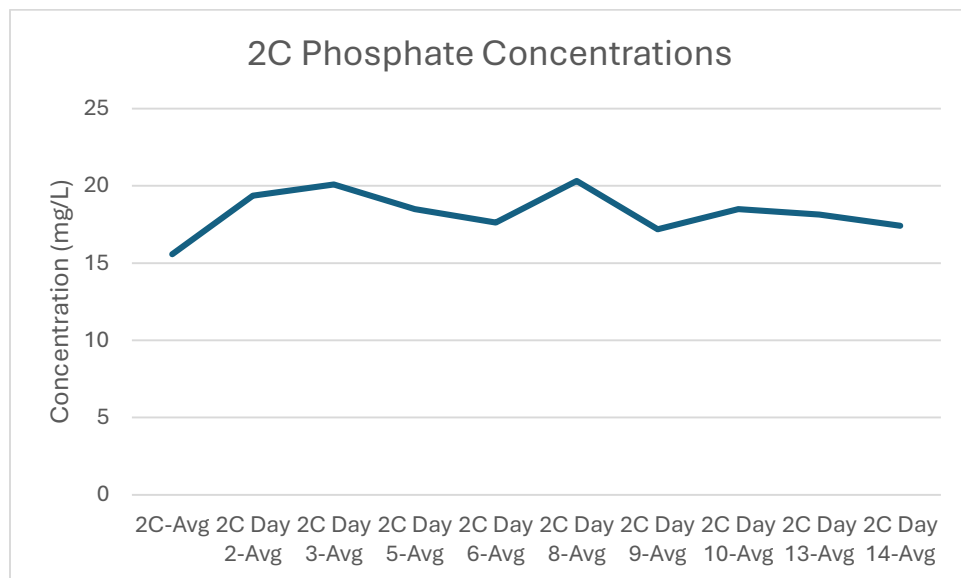
Figure 3*1C Phosphate Concentration over Time***Figure 4***2C Phosphate Concentration over Time*

Figure 5

3C Phosphate Concentration over Time

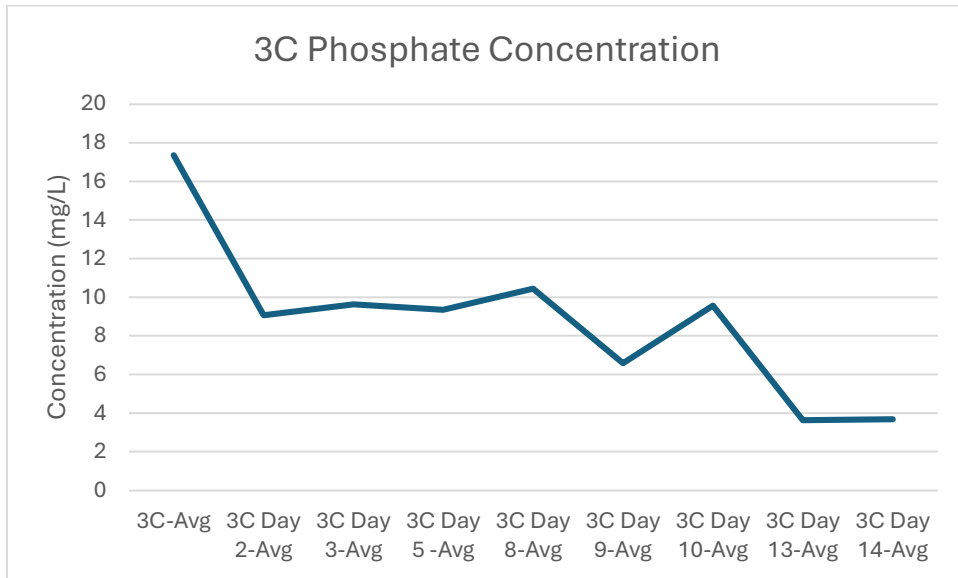


Figure 6

1D Phosphate Concentration over Time

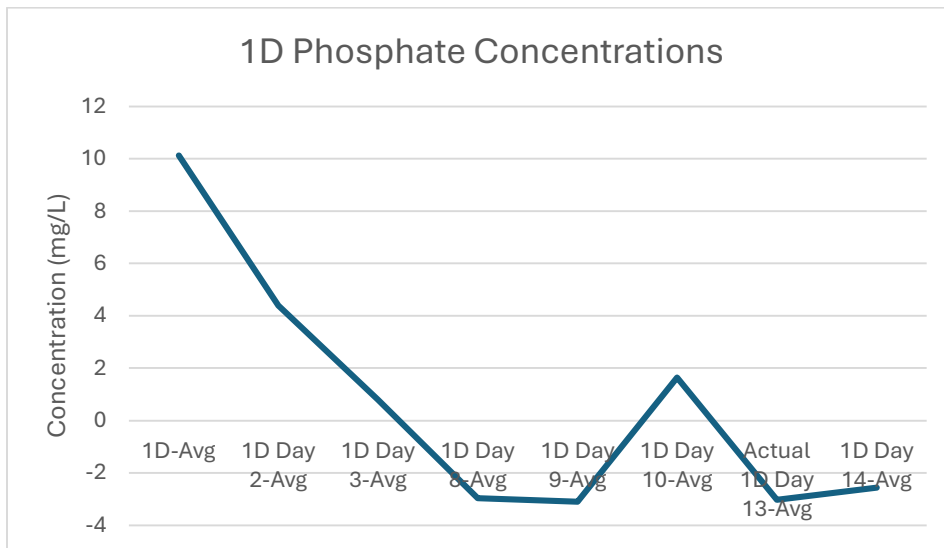


Figure 7

2D Phosphate Concentration over Time

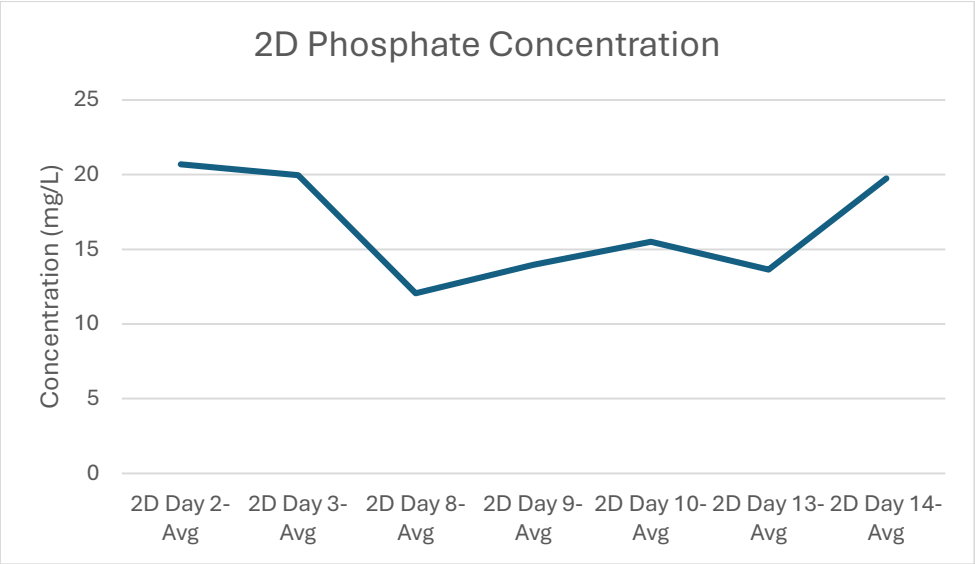


Figure 8

3D Phosphate Concentration over Time

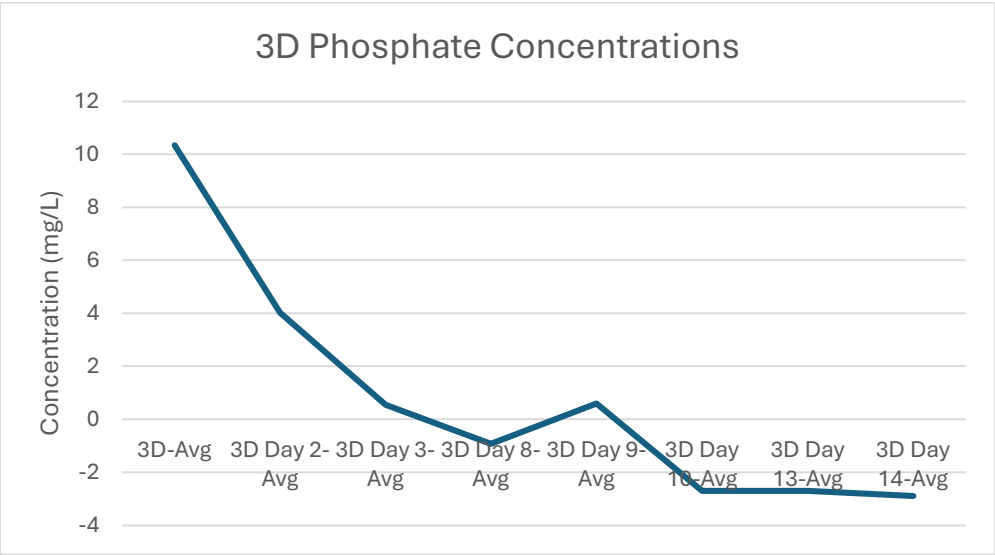


Figure 9

1C Nitrate Concentration over Time

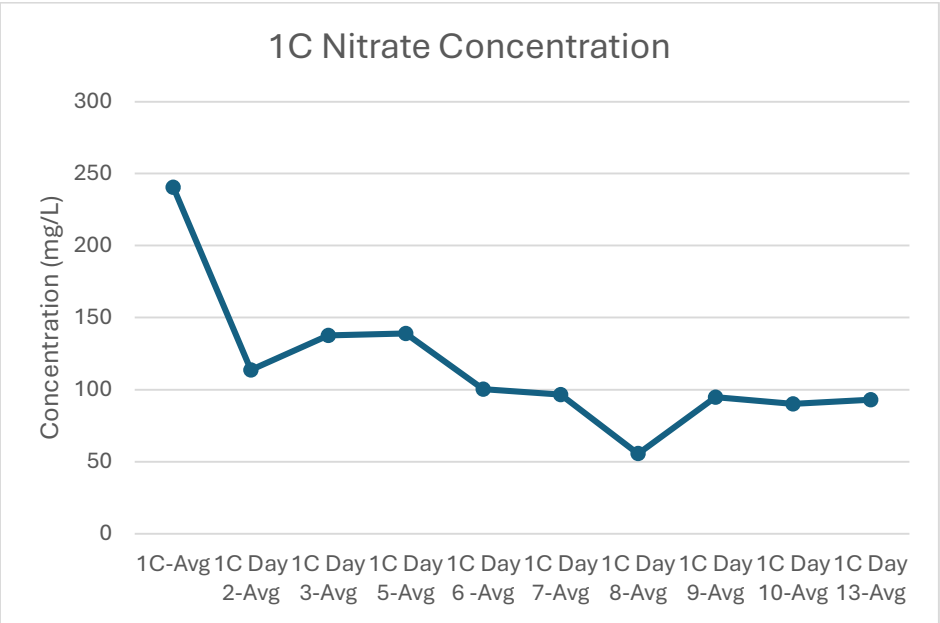


Figure 10

2C Nitrate Concentration over Time

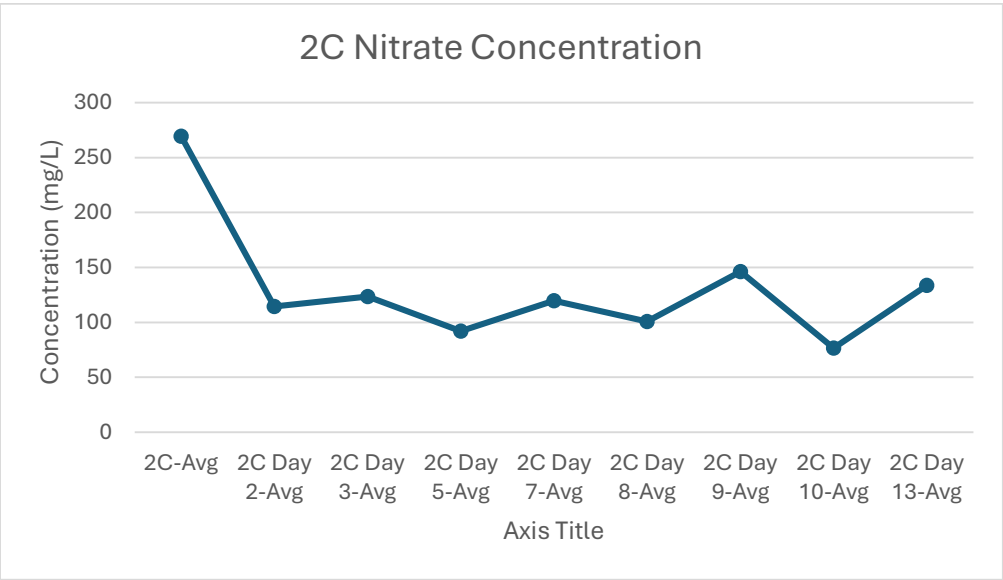


Figure 11

3C Nitrate Concentration over Time

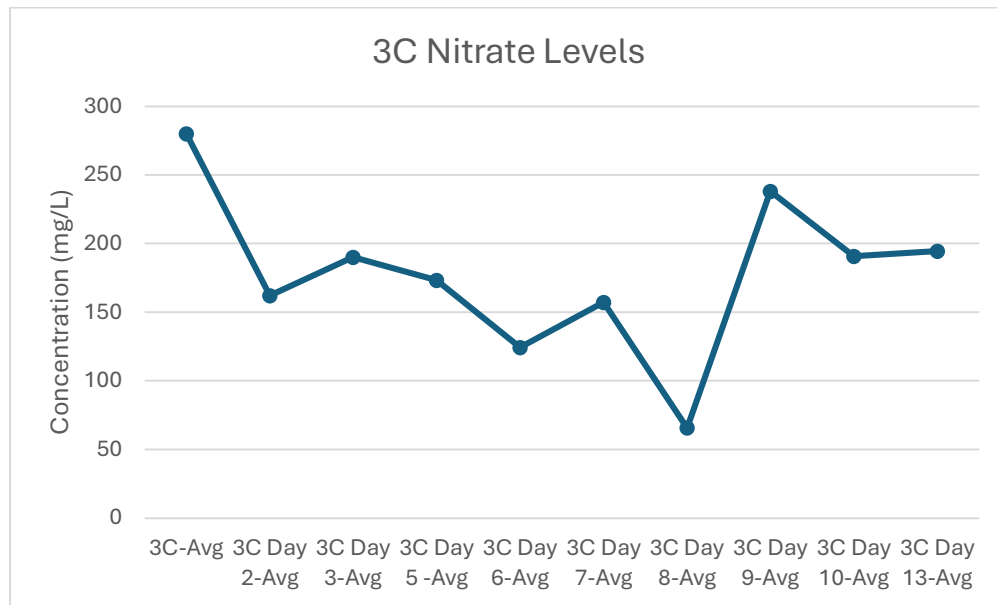


Figure 12

1D Nitrate Concentration over Time

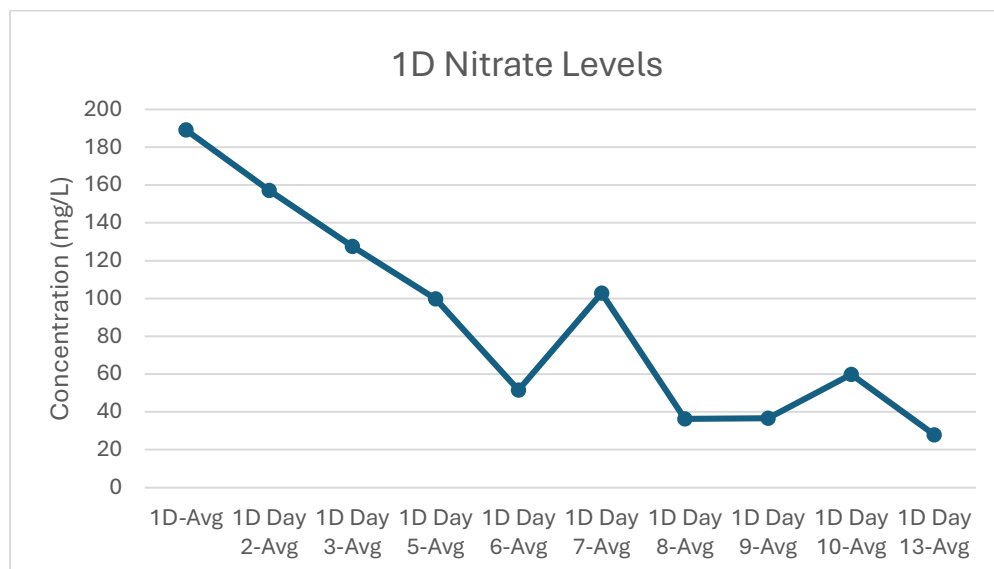
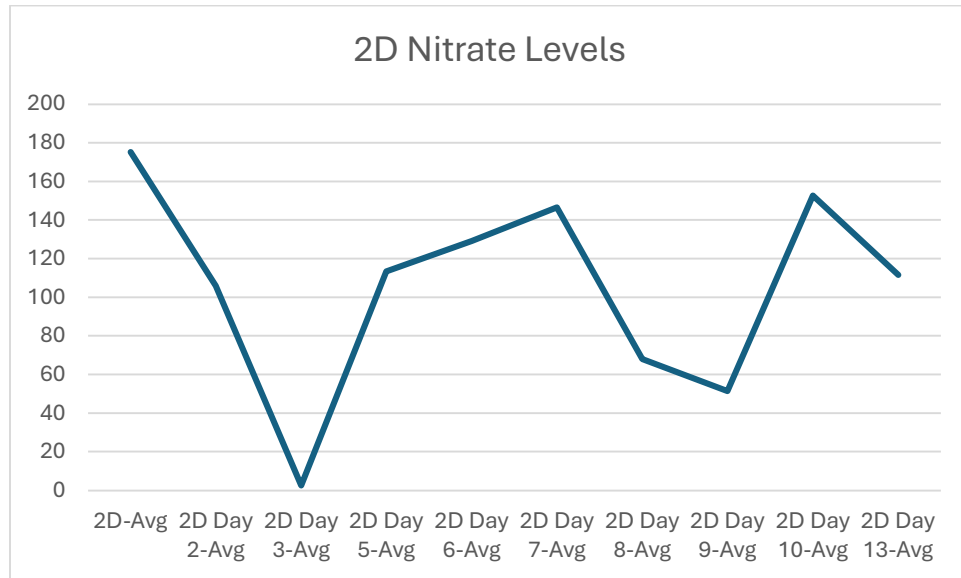
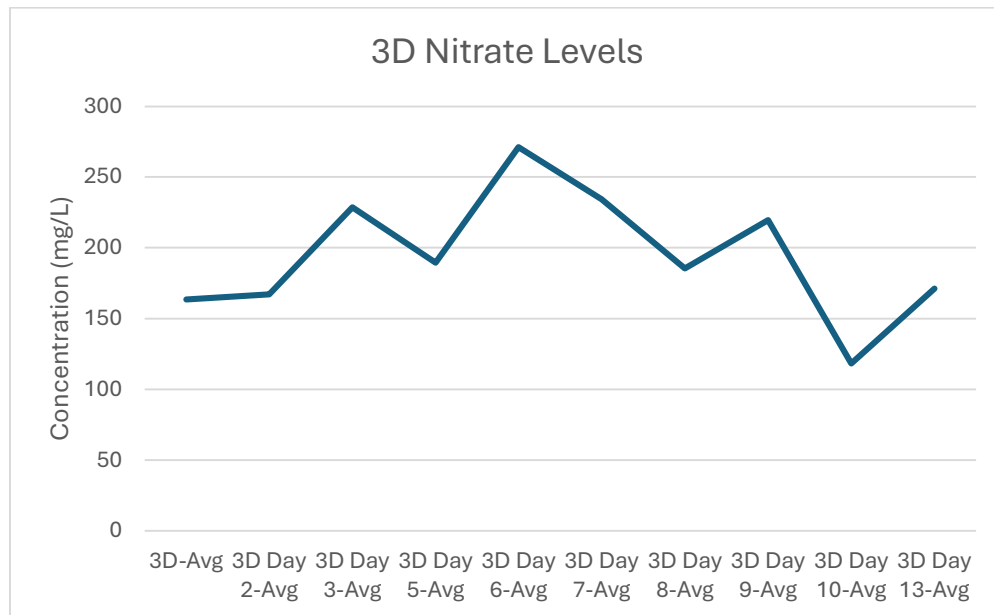


Figure 13*2D Nitrate Concentration over Time***Figure 14***3D Nitrate Concentration over Time*

Conclusion

From the results it can be concluded that in phosphate rich conditions *Desmodesmus Abundans* grows faster and denser than *Chlorella Vulgaris*. Between the two strains both nitrate heavy samples grew slower and less densely. There is also large inconsistency with the nitrate and phosphate concentration readings, and the method should be revisited to reduce the large amount of error seen.

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Appendices

Appendix A

S.O.P.O (Statement of Project Objectives)

- 1) Compare two algal strains *Chlorella Vulgaris* and *Desmodesmus Abundans* in three different conditions
- 2) Measure nitrate and phosphate levels of both the *Chlorella Vulgaris* and *Desmodesmus Abundans* while the strains are growing in the three different conditions

Appendix B

Methods

Preparation of the MLA medium

- 1) Pre-prepared media was ordered from [Chlorella fast grow nutrients mix - Maxpro Algae Thrive™ - LiveFishFood Australia](#)
- 2) 2g of MLA medium was added per 1L sterilized glass jar of deionized water

- 3) 1L jar with 2g MLA medium was shaken vigorously for 5 minutes to ensure even mixing

Preparation of the algae:

Labware:

- 6 1L jars
- 1 aquarium grade air pump
- 2 CO₂ pressurized cannisters
- 1 vacuum pump
- 1 set sterilized air lines
- 1 photospectrometer
- 12 filter papers

Desmodesmus Abundans

- 1) 3 1L Jars were obtained for the desmodesmus
- 2) 2g MLA medium/L added at about half a liter per jar (This is the control level of phosphate and nitrate)
- 3) Around 10 mL of the algae was then added to each jar
- 4) Samples were labeled 1D, 2D, 3D
 - a. 1D was the control group grown only with the 2g MLA media/L
 - b. 2D was the high phosphate with 100% increase
 - c. 3D was the high nitrate with 100% increase

- 5) The samples were then hooked up to an aquarium grade air pump to agitate the algae, the air was not inerted.
- 6) CO₂ was then slowly fed via pvc pipe for the rest of the experiment (21 days total)
- 7) 1mL was taken each day for cell counts

Chlorella Vulgaris

- 1) 2g MLA medium/L added to the three Chlorella Jars
- 2) Around 10 mL of the chlorella was then added
- 3) 100% increase in phosphate (0.0573 g KH₂PO₄ each) added to the 2D and 2C jars
- 4) 100% increase in Nitrate (0.3425 g NaNO₃ each) was added to the 3D and 3C jars
- 5) Samples were labeled
 - a. 1C was the control group made with 2g MLA media/L
 - b. 2C was the high phosphate group
 - c. 3C was the high nitrate group
- 6) The samples were then hooked up to an aquarium grade air pump to agitate the algae
- 7) Carbonation was then slowly fed via pvc pipe for the rest of the experiment (12 days total)
- 8) 5mL was taken each day for cell counts

UV Spectrophotometry

- 1) A series of known dilutions were made in order to first create a calibration curve so that concentration could be deduced based on absorptivity
- 2) First the sample was drawn out by a 5 mL syringe. 1mL of that sample was then added to 9mL of water through a syringe filter in order to eliminate the possibility of the algae's chlorophyll interfering with the UV spec results.
- 3) This dilution was then shaken gently to ensure even mixing.
- 4) 1 packet of the color changing substance was added
- 5) The mixture was shaken again to dissolve the powder.
- 6) This was repeated with all the samples (1D, 2D, 3D, 1C, 2C, 3C)
- 7) A Blank cuvette containing just deionized water was put through the UV Spectrometer
- 8) Once all samples were mixed, they were then added to a cuvette one by one to be put through the UV spectrometer
- 9) Data was saved and uploaded to a USB flash drive

Centrifuge

- 1) Vials of 50 mL were filled to be used in the centrifuge
- 2) An Allegra X-12 Centrifuge was used at a speed of 3,000 rpm
- 3) The top water was then decanted, while around 5mL was left to re-agitate the condensed algae pellet

- 4) This algae pellet was then added to the other centrifuged decanted and reagitated sample of the same type and concentration.
- 5) Each consolidated sample was then run another time through the centrifuge at 3,000 rpm

Dry Weight

- 1) 12 filter papers were weighed out (two for each sample) and then put on observation glasses labeled with the papers individual weight
- 2) Once all the samples had been taken for the UV Spec and Cell counts varying amounts were taken for filtering according to the table below
- 3) The filter pump was set up
- 4) Each sample was run through the filter pump on its own filter paper
- 5) The placement of which sample was on which paper was recorded throughout the process
- 6) The samples were then left out to dry overnight in the oven at 70 C 1 atm
- 7) The samples were then weighed out the following day once dry

Appendix C

Calculational Methods

Dry weight per volume

Filter paper weight was subtracted from the total dry weight to get the algae dry weight per volume

Total dry weight - Filter paper weight = Algae dry weight per volume

Appendix D

Raw Data

Figure D1.

Nitrate Levels Raw Data

Column1	Sample Name	Type	Conc	WL525.00	Result	Comments
1	1C	UNK-Rep		3.088		
2	1C-2	UNK-Rep		3.089		
3	1C-3	UNK-Rep		3.091		
4	1C-Avg	UNK-Avg	240.383	3.09	3.09	Avg of preceding 3 Samples
5	2C	UNK-Rep		3.361		
6	2C-2	UNK-Rep		3.363		
7	2C-3	UNK-Rep		3.361		
8	2C-Avg	UNK-Avg	269.494	3.362	3.362	Avg of preceding 3 Samples

9	3C	UNK-Rep		3.465		
10	3C-2	UNK-Rep		3.456		
11	3C-3	UNK-Rep		3.461		
12	3C-Avg	UNK-Avg	280.078	3.461	3.461	Avg of preceding 3 Samples
13	1D	UNK-Rep		2.611		
14	1D-2	UNK-Rep		2.611		
15	1D-3	UNK-Rep		2.608		
16	1D-Avg	UNK-Avg	189.081	2.61	2.61	Avg of preceding 3 Samples
17	2D	UNK-Rep		2.481		
18	2D-2	UNK-Rep		2.481		
19	2D-3	UNK-Rep		2.48		
20	2D-Avg	UNK-Avg	175.227	2.481	2.481	Avg of preceding 3 Samples
21	3D	UNK-Rep		2.371		
22	3D-2	UNK-Rep		2.37		
23	3D-3	UNK-Rep		2.371		
24	3D-Avg	UNK-Avg	163.456	2.371	2.371	Avg of preceding 3 Samples
25	1C Day 2	UNK-Rep		1.906		
26	1C Day 2-2	UNK-Rep		1.906		
27	1C Day 2-3	UNK-Rep		1.906		
28	1C Day 2-Avg	UNK-Avg	113.764	1.906	1.906	Avg of preceding 3 Samples
29	1D Day 2	UNK-Rep		2.311		
30	1D Day 2-2	UNK-Rep		2.311		
31	1D Day 2-3	UNK-Rep		2.312		

32	1D Day 2-Avg	UNK-Avg	157.136	2.311	2.311	Avg of preceding 3 Samples
33	2C Day 2	UNK-Rep		1.911		
34	2C Day 2-2	UNK-Rep		1.911		
35	2C Day 2-3	UNK-Rep		1.912		
36	2C Day 2-Avg	UNK-Avg	114.316	1.911	1.911	Avg of preceding 3 Samples
37	3D Day 2	UNK-Rep		2.405		
38	3D Day 2-2	UNK-Rep		2.404		
39	3D Day 2-3	UNK-Rep		2.405		
40	3D Day 2-Avg	UNK-Avg	167.099	2.405	2.405	Avg of preceding 3 Samples
41	2D Day 2	UNK-Rep		1.832		
42	2D Day 2-2	UNK-Rep		1.832		
43	2D Day 2-3	UNK-Rep		1.832		
44	2D Day 2-Avg	UNK-Avg	105.845	1.832	1.832	Avg of preceding 3 Samples
45	3C Day 2	UNK-Rep		2.358		
46	3C Day 2-2	UNK-Rep		2.358		
47	3C Day 2-3	UNK-Rep		2.357		
48	3C Day 2-Avg	UNK-Avg	162.083	2.358	2.358	Avg of preceding 3 Samples
49	1C Day 3	UNK-Rep		2.131		
50	1C Day 3-2	UNK-Rep		2.13		
51	1C Day 3-3	UNK-Rep		2.131		
52	1C Day 3-Avg	UNK-Avg	137.776	2.13	2.13	Avg of preceding 3 Samples
53	1D Day 3	UNK-Rep		2.035		
54	1D Day 3-2	UNK-Rep		2.035		

55	1D Day 3-3	UNK-Rep		2.035		
56	1D Day 3-Avg	UNK-Avg	127.581	2.035	2.035	Avg of preceding 3 Samples
57	2C Day 3	UNK-Rep		1.995		
58	2C Day 3-2	UNK-Rep		1.996		
59	2C Day 3-3	UNK-Rep		1.995		
60	2C Day 3-Avg	UNK-Avg	123.341	1.996	1.996	Avg of preceding 3 Samples
61	2D Day 3	UNK-Rep		0.867		
62	2D Day 3-2	UNK-Rep		0.867		
63	2D Day 3-3	UNK-Rep		0.867		
64	2D Day 3-Avg	UNK-Avg	2.618	0.867	0.867	Avg of preceding 3 Samples
65	3C Day 3	UNK-Rep		2.619		
66	3C Day 3-2	UNK-Rep		2.621		
67	3C Day 3-3	UNK-Rep		2.621		
68	3C Day 3-Avg	UNK-Avg	190.185	2.62	2.62	Avg of preceding 3 Samples
69	3D Day 3	UNK-Rep		2.98		
70	3D Day 3-2	UNK-Rep		2.979		
71	3D Day 3-3	UNK-Rep		2.978		
72	3D Day 3-Avg	UNK-Avg	228.537	2.979	2.979	Avg of preceding 3 Samples
73	1C Day 5	UNK-Rep		2.143		
74	1C Day 5-2	UNK-Rep		2.141		
75	1C Day 5-3	UNK-Rep		2.142		
76	1C Day 5-Avg	UNK-Avg	139.007	2.142	2.142	Avg of preceding 3 Samples
77	2C Day 5	UNK-Rep		1.703		

78	2C Day 5-2	UNK-Rep		1.704		
79	2C Day 5-3	UNK-Rep		1.703		
80	2C Day 5-Avg	UNK-Avg	92.092	1.704	1.704	Avg of preceding 3 Samples
81	3C Day 5	UNK-Rep		2.461		
82	3C Day 5 -2	UNK-Rep		2.461		
83	3C Day 5 -3	UNK-Rep		2.463		
84	3C Day 5 -Avg	UNK-Avg	173.169	2.461	2.461	Avg of preceding 3 Samples
85	1C Day 6	UNK-Rep		1.78		
86	1C Day 6 -2	UNK-Rep		1.781		
87	1C Day 6 -3	UNK-Rep		1.781		
88	1C Day 6 -Avg	UNK-Avg	100.327	1.78	1.78	Avg of preceding 3 Samples
89	2C Day 6	UNK-Rep		1.532		
90	2C Day 6-2	UNK-Rep		1.532		
91	2C Day 6-3	UNK-Rep		1.532		
92	2C Day 6-Avg	UNK-Avg	73.724	1.532	1.532	Avg of preceding 3 Samples
93	3C Day 6	UNK-Rep		2.005		
94	3C Day 6-2	UNK-Rep		2.005		
95	3C Day 6-3	UNK-Rep		2.006		
96	3C Day 6-Avg	UNK-Avg	124.412	2.006	2.006	Avg of preceding 3 Samples
97	1C Day 7	UNK-Rep		1.745		
98	1C Day 7-2	UNK-Rep		1.745		
99	1C Day 7-3	UNK-Rep		1.746		
100	1C Day 7-Avg	UNK-Avg	96.57	1.745	1.745	Avg of preceding 3 Samples

101	2C Day 7	UNK-Rep	1.961			
102	2C Day 7-2	UNK-Rep	1.962			
103	2C Day 7-3	UNK-Rep	1.963			
104	2C Day 7-Avg	UNK-Avg	119.748	1.962	1.962	Avg of preceding 3 Samples
105	3C Day 7	UNK-Rep	2.311			
106	3C Day 7-2	UNK-Rep	2.312			
107	3C Day 7-3	UNK-Rep	2.312			
108	3C Day 7-Avg	UNK-Avg	157.204	2.312	2.312	Avg of preceding 3 Samples
109	1D Day 5	UNK-Rep	1.776			
110	1D Day 5-2	UNK-Rep	1.776			
111	1D Day 5-3	UNK-Rep	1.776			
112	1D Day 5-Avg	UNK-Avg	99.805	1.776	1.776	Avg of preceding 3 Samples
113	2D Day 5	UNK-Rep	1.903			
114	2D Day 5-2	UNK-Rep	1.903			
115	2D Day 5-3	UNK-Rep	1.903			
116	2D Day 5-Avg	UNK-Avg	113.41	1.903	1.903	Avg of preceding 3 Samples
117	3D Day 5	UNK-Rep	2.613			
118	3D Day 5-2	UNK-Rep	2.614			
119	3D Day 5-3	UNK-Rep	2.614			
120	3D Day 5-Avg	UNK-Avg	189.491	2.614	2.614	Avg of preceding 3 Samples
121	1D Day 6	UNK-Rep	1.324			
122	1D Day 6-2	UNK-Rep	1.324			
123	1D Day 6-3	UNK-Rep	1.324			

124	1D Day 6-Avg	UNK-Avg	51.51	1.324	1.324	Avg of preceding 3 Samples
125	2D Day 6	UNK-Rep		2.049		
126	2D Day 6-2	UNK-Rep		2.049		
127	2D Day 6-3	UNK-Rep		2.049		
128	2D Day 6-Avg	UNK-Avg	129.07	2.049	2.049	Avg of preceding 3 Samples
129	3D Day 6	UNK-Rep		3.377		
130	3D Day 6-2	UNK-Rep		3.376		
131	3D Day 6-3	UNK-Rep		3.378		
132	3D Day 6-Avg	UNK-Avg	271.159	3.377	3.377	Avg of preceding 3 Samples
133	1D Day 7	UNK-Rep		1.805		
134	1D Day 7-2	UNK-Rep		1.804		
135	1D Day 7-3	UNK-Rep		1.805		
136	1D Day 7-Avg	UNK-Avg	102.909	1.805	1.805	Avg of preceding 3 Samples
137	2D Day 7	UNK-Rep		2.212		
138	2D Day 7-2	UNK-Rep		2.212		
139	2D Day 7-3	UNK-Rep		2.212		
140	2D Day 7-Avg	UNK-Avg	146.509	2.212	2.212	Avg of preceding 3 Samples
141	3D Day 7	UNK-Rep		3.032		
142	3D Day 7-2	UNK-Rep		3.033		
143	3D Day 7-3	UNK-Rep		3.033		
144	3D Day 7-Avg	UNK-Avg	234.257	3.032	3.032	Avg of preceding 3 Samples
145	1C Day 6 re-eval	UNK-Rep		1.753		
146	1C Day 6 re-eval-2	UNK-Rep		1.753		

147	1C Day 6 re-eval- 3	UNK- Rep		1.753		
148	1C Day 6 re-eval- Avg	UNK- Avg	97.4	1.753	1.753	Avg of preceding 3 Samples
149	2C Day 6 re-eval	UNK- Rep		1.853		
150	2C Day 6 re-eval- 2	UNK- Rep		1.853		
151	2C Day 6 re-eval- 3	UNK- Rep		1.853		
152	2C Day 6 re-eval- Avg	UNK- Avg	108.082	1.853	1.853	Avg of preceding 3 Samples
153	3C Day 6 re-eval	UNK- Rep		2.299		
154	3C Day 6 re-eval- 2	UNK- Rep		2.298		
155	3C Day 6 re-eval- 3	UNK- Rep		2.3		
156	3C Day 6 re-eval- Avg	UNK- Avg	155.806	2.299	2.299	Avg of preceding 3 Samples
157	1D Day 6 re-eval	UNK- Rep		1.184		
158	1D Day 6 re-eval- 2	UNK- Rep		1.184		
159	1D Day 6 re-eval- 3	UNK- Rep		1.183		
160	1D Day 6 re-eval- Avg	UNK- Avg	36.463	1.184	1.184	Avg of preceding 3 Samples
161	2D Day 6 re-eval	UNK- Rep		1.69		
162	2D Day 6 re-eval- 2	UNK- Rep		1.69		
163	2D Day 6 re-eval- 3	UNK- Rep		1.689		
164	2D Day 6 re-eval- Avg	UNK- Avg	90.587	1.689	1.689	Avg of preceding 3 Samples
165	3D Day 6 re-eval	UNK- Rep		2.17		
166	3D Day 6 re-eval- 2	UNK- Rep		2.171		
167	3D Day 6 re-eval- 3	UNK- Rep		2.17		
168	3D Day 6 re-eval- Avg	UNK- Avg	142.031	2.17	2.17	Avg of preceding 3 Samples
169	1C Day 8	UNK- Rep		1.363		

170	1C Day 8-2	UNK-Rep		1.363		
171	1C Day 8-3	UNK-Rep		1.364		
172	1C Day 8-Avg	UNK-Avg	55.696	1.363	1.363	Avg of preceding 3 Samples
173	2C Day 8	UNK-Rep		1.783		
174	2C Day 8-2	UNK-Rep		1.784		
175	2C Day 8-3	UNK-Rep		1.784		
176	2C Day 8-Avg	UNK-Avg	100.686	1.784	1.784	Avg of preceding 3 Samples
177	3C Day 8	UNK-Rep		1.456		
178	3C Day 8-2	UNK-Rep		1.457		
179	3C Day 8-3	UNK-Rep		1.459		
180	3C Day 8-Avg	UNK-Avg	65.774	1.458	1.458	Avg of preceding 3 Samples
181	1D Day 8	UNK-Rep		1.182		
182	1D Day 8-2	UNK-Rep		1.181		
183	1D Day 8-3	UNK-Rep		1.182		
184	1D Day 8-Avg	UNK-Avg	36.24	1.181	1.181	Avg of preceding 3 Samples
185	2D Day 8	UNK-Rep		1.477		
186	2D Day 8-2	UNK-Rep		1.477		
187	2D Day 8-3	UNK-Rep		1.477		
188	2D Day 8-Avg	UNK-Avg	67.885	1.477	1.477	Avg of preceding 3 Samples
189	3D Day 8	UNK-Rep		2.575		
190	3D Day 8-2	UNK-Rep		2.574		
191	3D Day 8-3	UNK-Rep		2.575		
192	3D Day 8-Avg	UNK-Avg	185.312	2.575	2.575	Avg of preceding 3 Samples

193	1C Day 9	UNK-Rep		1.728		
194	1C Day 9-2	UNK-Rep		1.729		
195	1C Day 9-3	UNK-Rep		1.73		
196	1C Day 9-Avg	UNK-Avg	94.822	1.729	1.729	Avg of preceding 3 Samples
197	2C Day 9	UNK-Rep		2.207		
198	2C Day 9-2	UNK-Rep		2.209		
199	2C Day 9-3	UNK-Rep		2.208		
200	2C Day 9-Avg	UNK-Avg	146.044	2.208	2.208	Avg of preceding 3 Samples
201	3C Day 9	UNK-Rep		3.069		
202	3C Day 9-2	UNK-Rep		3.071		
203	3C Day 9-3	UNK-Rep		3.067		
204	3C Day 9-Avg	UNK-Avg	238.197	3.069	3.069	Avg of preceding 3 Samples
205	1D Day 9	UNK-Rep		1.185		
206	1D Day 9-2	UNK-Rep		1.184		
207	1D Day 9-3	UNK-Rep		1.184		
208	1D Day 9-Avg	UNK-Avg	36.572	1.185	1.185	Avg of preceding 3 Samples
209	2D Day 9	UNK-Rep		1.323		
210	2D Day 9-2	UNK-Rep		1.323		
211	2D Day 9-3	UNK-Rep		1.323		
212	2D Day 9-Avg	UNK-Avg	51.367	1.323	1.323	Avg of preceding 3 Samples
213	3D Day 9	UNK-Rep		2.895		
214	3D Day 9-2	UNK-Rep		2.895		
215	3D Day 9-3	UNK-Rep		2.894		

216	3D Day 9-Avg	UNK-Avg	219.539	2.895	2.895	Avg of preceding 3 Samples
217	1D Day 10	UNK-Rep		1.399		
218	1D Day 10-2	UNK-Rep		1.401		
219	1D Day 10-3	UNK-Rep		1.403		
220	1D Day 10-Avg	UNK-Avg	59.758	1.401	1.401	Avg of preceding 3 Samples
221	2D Day 10	UNK-Rep		2.269		
222	2D Day 10-2	UNK-Rep		2.269		
223	2D Day 10-3	UNK-Rep		2.269		
224	2D Day 10-Avg	UNK-Avg	152.578	2.269	2.269	Avg of preceding 3 Samples
225	3D Day 10	UNK-Rep		1.949		
226	3D Day 10-2	UNK-Rep		1.949		
227	3D Day 10-3	UNK-Rep		1.951		
228	3D Day 10-Avg	UNK-Avg	118.441	1.95	1.95	Avg of preceding 3 Samples
229	1C Day 10	UNK-Rep		1.686		
230	1C Day 10-2	UNK-Rep		1.686		
231	1C Day 10-3	UNK-Rep		1.687		
232	1C Day 10-Avg	UNK-Avg	90.227	1.686	1.686	Avg of preceding 3 Samples
233	2C Day 10	UNK-Rep		1.559		
234	2C Day 10-2	UNK-Rep		1.559		
235	2C Day 10-3	UNK-Rep		1.559		
236	2C Day 10-Avg	UNK-Avg	76.644	1.559	1.559	Avg of preceding 3 Samples
237	3C Day 10	UNK-Rep		2.626		
238	3C Day 10-2	UNK-Rep		2.624		

239	3C Day 10-3	UNK-Rep		2.625		
240	3C Day 10-Avg	UNK-Avg	190.66	2.625	2.625	Avg of preceding 3 Samples
241	1D Day 13	UNK-Rep		1.104		
242	1D Day 13-2	UNK-Rep		1.104		
243	1D Day 13-3	UNK-Rep		1.104		
244	1D Day 13-Avg	UNK-Avg	27.933	1.104	1.104	Avg of preceding 3 Samples
245	1C Day 13	UNK-Rep		1.713		
246	1C Day 13-2	UNK-Rep		1.713		
247	1C Day 13-3	UNK-Rep		1.713		
248	1C Day 13-Avg	UNK-Avg	93.073	1.713	1.713	Avg of preceding 3 Samples
249	2D Day 13	UNK-Rep		1.886		
250	2D Day 13-2	UNK-Rep		1.886		
251	2D Day 13-3	UNK-Rep		1.887		
252	2D Day 13-Avg	UNK-Avg	111.627	1.886	1.886	Avg of preceding 3 Samples
253	2C Day 13	UNK-Rep		2.091		
254	2C Day 13-2	UNK-Rep		2.09		
255	2C Day 13-3	UNK-Rep		2.091		
256	2C Day 13-Avg	UNK-Avg	133.55	2.091	2.091	Avg of preceding 3 Samples
257	3D Day 13	UNK-Rep		2.443		
258	3D Day 13-2	UNK-Rep		2.443		
259	3D Day 13-3	UNK-Rep		2.443		
260	3D Day 13-Avg	UNK-Avg	171.228	2.443	2.443	Avg of preceding 3 Samples
261	3C Day 13	UNK-Rep		2.66		

262	3C Day 13-2	UNK-Rep		2.661		
263	3C Day 13-3	UNK-Rep		2.66		
264	3C Day 13-Avg	UNK-Avg	194.444	2.66	2.66	Avg of preceding 3 Samples

Figure D2.*Phosphate Concentrations Raw Data*

Column1	Sample Name	Type	Conc	WL525.00	Result	Comments
1	1C	UNK-Rep		0.541		
2	1C-2	UNK-Rep		0.54		
3	1C-3	UNK-Rep		0.54		
4	1C-Avg	UNK-Avg	17.272	0.541	0.541	Avg of preceding 3 Samples
5	2C	UNK-Rep		0.497		
6	2C-2	UNK-Rep		0.497		
7	2C-3	UNK-Rep		0.497		
8	2C-Avg	UNK-Avg	15.583	0.497	0.497	Avg of preceding 3 Samples
9	3C	UNK-Rep		0.543		
10	3C-2	UNK-Rep		0.543		
11	3C-3	UNK-Rep		0.543		
12	3C-Avg	UNK-Avg	17.359	0.543	0.543	Avg of preceding 3 Samples
13	1D	UNK-Rep		0.356		
14	1D-2	UNK-Rep		0.356		
15	1D-3	UNK-Rep		0.356		
16	1D-Avg	UNK-Avg	10.131	0.356	0.356	Avg of preceding 3 Samples

17		UNK-Rep		0.298		
18		UNK-Rep		0.298		
19		UNK-Rep		0.298		
20	2D	UNK-Avg	7.87	0.298	0.298	Avg of preceding 3 Samples
21	3D	UNK-Rep		0.362		
22	3D-2	UNK-Rep		0.362		
23	3D-3	UNK-Rep		0.362		
24	3D-Avg	UNK-Avg	10.343	0.362	0.362	Avg of preceding 3 Samples
25	1C Day 2	UNK-Rep		0.338		
26	1C Day 2-2	UNK-Rep		0.338		
27	1C Day 2-3	UNK-Rep		0.338		
28	1C Day 2-Avg	UNK-Avg	9.413	0.338	0.338	Avg of preceding 3 Samples
29	1D Day 2	UNK-Rep		0.208		
30	1D Day 2-2	UNK-Rep		0.208		
31	1D Day 2-3	UNK-Rep		0.208		
32	1D Day 2-Avg	UNK-Avg	4.384	0.208	0.208	Avg of preceding 3 Samples
33	2C Day 2	UNK-Rep		0.594		
34	2C Day 2-2	UNK-Rep		0.594		
35	2C Day 2-3	UNK-Rep		0.594		
36	2C Day 2-Avg	UNK-Avg	19.359	0.594	0.594	Avg of preceding 3 Samples
37	2D Day 2	UNK-Rep		0.628		
38	2D Day 2-2	UNK-Rep		0.629		
39	2D Day 2-3	UNK-Rep		0.629		

40	2D Day 2-Avg	UNK-Avg	20.693	0.629	0.629	Avg of preceding 3 Samples
41	3C Day 2	UNK-Rep		0.329		
42	3C Day 2-2	UNK-Rep		0.329		
43	3C Day 2-3	UNK-Rep		0.329		
44	3C Day 2-Avg	UNK-Avg	9.07	0.329	0.329	Avg of preceding 3 Samples
45	3D Day 2	UNK-Rep		0.199		
46	3D Day 2-2	UNK-Rep		0.199		
47	3D Day 2-3	UNK-Rep		0.199		
48	3D Day 2-Avg	UNK-Avg	4.028	0.199	0.199	Avg of preceding 3 Samples
49	1C Day 3	UNK-Rep		0.229		
50	1C Day 3-2	UNK-Rep		0.229		
51	1C Day 3-3	UNK-Rep		0.229		
52	1C Day 3-Avg	UNK-Avg	5.191	0.229	0.229	Avg of preceding 3 Samples
53	1D Day 3	UNK-Rep		0.115		
54	1D Day 3-2	UNK-Rep		0.115		
55	1D Day 3-3	UNK-Rep		0.115		
56	1D Day 3-Avg	UNK-Avg	0.794	0.115	0.115	Avg of preceding 3 Samples
57	2C Day 3	UNK-Rep		0.613		
58	2C Day 3-2	UNK-Rep		0.613		
59	2C Day 3-3	UNK-Rep		0.613		
60	2C Day 3-Avg	UNK-Avg	20.085	0.613	0.613	Avg of preceding 3 Samples
61	2D Day 3	UNK-Rep		0.61		
62	2D Day 3-2	UNK-Rep		0.61		

63	2D Day 3-3	UNK-Rep		0.61		
64	2D Day 3-Avg	UNK-Avg	19.954	0.61	0.61	Avg of preceding 3 Samples
65	3C Day 3	UNK-Rep		0.344		
66	3C Day 3-2	UNK-Rep		0.344		
67	3C Day 3-3	UNK-Rep		0.344		
68	3C Day 3-Avg	UNK-Avg	9.639	0.344	0.344	Avg of preceding 3 Samples
69	3D Day 3	UNK-Rep		0.109		
70	3D Day 3-2	UNK-Rep		0.109		
71	3D Day 3-3	UNK-Rep		0.109		
72	3D Day 3-Avg	UNK-Avg	0.548	0.109	0.109	Avg of preceding 3 Samples
73	1C Day 5	UNK-Rep		0.128		
74	1C Day 5-2	UNK-Rep		0.128		
75	1C Day 5-3	UNK-Rep		0.128		
76	1C Day 5-Avg	UNK-Avg	1.276	0.128	0.128	Avg of preceding 3 Samples
77	2C Day 5	UNK-Rep		0.572		
78	2C Day 5-2	UNK-Rep		0.572		
79	2C Day 5-3	UNK-Rep		0.572		
80	2C Day 5-Avg	UNK-Avg	18.493	0.572	0.572	Avg of preceding 3 Samples
81	3C Day 5	UNK-Rep		0.336		
82	3C Day 5 -2	UNK-Rep		0.336		
83	3C Day 5 -3	UNK-Rep		0.336		
84	3C Day 5 -Avg	UNK-Avg	9.357	0.336	0.336	Avg of preceding 3 Samples
85	1C Day 6	UNK-Rep		0.166		

86	1C Day 6-2	UNK-Rep		0.166		
87	1C Day 6-3	UNK-Rep		0.166		
88	1C Day 6-Avg	UNK-Avg	2.77	0.166	0.166	Avg of preceding 3 Samples
89	2C Day 6	UNK-Rep		0.549		
90	2C Day 6-2	UNK-Rep		0.549		
91	2C Day 6-3	UNK-Rep		0.549		
92	2C Day 6-Avg	UNK-Avg	17.619	0.549	0.549	Avg of preceding 3 Samples
93	1C Day 8	UNK-Rep		0.116		
94	1C Day 8-2	UNK-Rep		0.116		
95	1C Day 8-3	UNK-Rep		0.116		
96	1C Day 8-Avg	UNK-Avg	0.819	0.116	0.116	Avg of preceding 3 Samples
97	2C Day 8	UNK-Rep		0.619		
98	2C Day 8-2	UNK-Rep		0.619		
99	2C Day 8-3	UNK-Rep		0.619		
100	2C Day 8-Avg	UNK-Avg	20.329	0.619	0.619	Avg of preceding 3 Samples
101	3C Day 8	UNK-Rep		0.364		
102	3C Day 8-2	UNK-Rep		0.364		
103	3C Day 8-3	UNK-Rep		0.364		
104	3C Day 8-Avg	UNK-Avg	10.439	0.364	0.364	Avg of preceding 3 Samples
105	1D Day 8	UNK-Rep		0.018		
106	1D Day 8-2	UNK-Rep		0.018		
107	1D Day 8-3	UNK-Rep		0.018		
108	1D Day 8-Avg	UNK-Avg	-2.966	0.018	0.018	Avg of preceding 3 Samples

109	2D Day 8	UNK-Rep		0.406		
110	2D Day 8-2	UNK-Rep		0.406		
111	2D Day 8-3	UNK-Rep		0.406		
112	2D Day 8-Avg	UNK-Avg	12.06	0.406	0.406	Avg of preceding 3 Samples
113	3D Day 8	UNK-Rep		0.071		
114	3D Day 8-2	UNK-Rep		0.071		
115	3D Day 8-3	UNK-Rep		0.071		
116	3D Day 8-Avg	UNK-Avg	-0.932	0.071	0.071	Avg of preceding 3 Samples
117	1D Day 9	UNK-Rep		0.015		
118	1D Day 9-2	UNK-Rep		0.015		
119	1D Day 9-3	UNK-Rep		0.015		
120	1D Day 9-Avg	UNK-Avg	-3.095	0.015	0.015	Avg of preceding 3 Samples
121	2D Day 9	UNK-Rep		0.455		
122	2D Day 9-2	UNK-Rep		0.455		
123	2D Day 9-3	UNK-Rep		0.455		
124	2D Day 9-Avg	UNK-Avg	13.968	0.455	0.455	Avg of preceding 3 Samples
125	3D Day 9	UNK-Rep		0.11		
126	3D Day 9-2	UNK-Rep		0.11		
127	3D Day 9-3	UNK-Rep		0.11		
128	3D Day 9-Avg	UNK-Avg	0.594	0.11	0.11	Avg of preceding 3 Samples
129	1C Day 9	UNK-Rep		0.107		
130	1C Day 9-2	UNK-Rep		0.107		
131	1C Day 9-3	UNK-Rep		0.107		

132	1C Day 9-Avg	UNK-Avg	0.478	0.107	0.107	Avg of preceding 3 Samples
133	2C Day 9	UNK-Rep		0.538		
134	2C Day 9-2	UNK-Rep		0.539		
135	2C Day 9-3	UNK-Rep		0.538		
136	2C Day 9-Avg	UNK-Avg	17.191	0.538	0.538	Avg of preceding 3 Samples
137	3C Day 9	UNK-Rep		0.265		
138	3C Day 9-2	UNK-Rep		0.265		
139	3C Day 9-3	UNK-Rep		0.265		
140	3C Day 9-Avg	UNK-Avg	6.579	0.265	0.265	Avg of preceding 3 Samples
141	1C Day 10	UNK-Rep		0.117		
142	1C Day 10-2	UNK-Rep		0.117		
143	1C Day 10-3	UNK-Rep		0.117		
144	1C Day 10-Avg	UNK-Avg	0.863	0.117	0.117	Avg of preceding 3 Samples
145	2C Day 10	UNK-Rep		0.572		
146	2C Day 10-2	UNK-Rep		0.572		
147	2C Day 10-3	UNK-Rep		0.572		
148	2C Day 10-Avg	UNK-Avg	18.484	0.572	0.572	Avg of preceding 3 Samples
149	3C Day 10	UNK-Rep		0.342		
150	3C Day 10-2	UNK-Rep		0.342		
151	3C Day 10-3	UNK-Rep		0.342		
152	3C Day 10-Avg	UNK-Avg	9.569	0.342	0.342	Avg of preceding 3 Samples
153	1D Day 10	UNK-Rep		0.137		
154	1D Day 10-2	UNK-Rep		0.137		

155	1D Day 10-3	UNK-Rep		0.138		
156	1D Day 10-Avg	UNK-Avg	1.638	0.137	0.137	Avg of preceding 3 Samples
157	2D Day 10	UNK-Rep		0.495		
158	2D Day 10-2	UNK-Rep		0.495		
159	2D Day 10-3	UNK-Rep		0.495		
160	2D Day 10-Avg	UNK-Avg	15.497	0.495	0.495	Avg of preceding 3 Samples
161	3D Day 10	UNK-Rep		0.025		
162	3D Day 10-2	UNK-Rep		0.025		
163	3D Day 10-3	UNK-Rep		0.025		
164	3D Day 10-Avg	UNK-Avg	-2.704	0.025	0.025	Avg of preceding 3 Samples
165	1D Day 13	UNK-Rep		0.022		
166	1D Day 13-2	UNK-Rep		0.022		
167	1D Day 13-3	UNK-Rep		0.022		
168	1D Day 13-Avg	UNK-Avg	-2.843	0.022	0.022	Avg of preceding 3 Samples
169	Actual 1D Day 13	UNK-Rep		0.017		
170	Actual 1D Day 13-2	UNK-Rep		0.017		
171	Actual 1D Day 13-3	UNK-Rep		0.017		
172	Actual 1D Day 13-Avg	UNK-Avg	-3.025	0.017	0.017	Avg of preceding 3 Samples
173	1C Day 13	UNK-Rep		0.089		
174	1C Day 13-2	UNK-Rep		0.089		
175	1C Day 13-3	UNK-Rep		0.089		
176	1C Day 13-Avg	UNK-Avg	-0.231	0.089	0.089	Avg of preceding 3 Samples
177	2D Day 13	UNK-Rep		0.447		

178	2D Day 13-2	UNK-Rep	0.447			
179	2D Day 13-3	UNK-Rep	0.447			
180	2D Day 13-Avg	UNK-Avg	13.646	0.447	0.447	Avg of preceding 3 Samples
181	2C Day 13	UNK-Rep	0.563			
182	2C Day 13-2	UNK-Rep	0.563			
183	2C Day 13-3	UNK-Rep	0.563			
184	2C Day 13-Avg	UNK-Avg	18.146	0.563	0.563	Avg of preceding 3 Samples
185	3D Day 13	UNK-Rep	0.025			
186	3D Day 13-2	UNK-Rep	0.025			
187	3D Day 13-3	UNK-Rep	0.025			
188	3D Day 13-Avg	UNK-Avg	-2.704	0.025	0.025	Avg of preceding 3 Samples
189	3C Day 13	UNK-Rep	0.189			
190	3C Day 13-2	UNK-Rep	0.189			
191	3C Day 13-3	UNK-Rep	0.189			
192	3C Day 13-Avg	UNK-Avg	3.634	0.189	0.189	Avg of preceding 3 Samples
193	1C Day 14	UNK-Rep	0.05			
194	1C Day 14-2	UNK-Rep	0.05			
195	1C Day 14-3	UNK-Rep	0.05			
196	1C Day 14-Avg	UNK-Avg	-1.751	0.05	0.05	Avg of preceding 3 Samples
197	1D Day 14	UNK-Rep	0.029			
198	1D Day 14-2	UNK-Rep	0.029			
199	1D Day 14-3	UNK-Rep	0.029			
200	1D Day 14-Avg	UNK-Avg	-2.565	0.029	0.029	Avg of preceding 3 Samples

201	2C Day 14	UNK-Rep		0.544		
202	2C Day 14-2	UNK-Rep		0.544		
203	2C Day 14-3	UNK-Rep		0.544		
204	2C Day 14-Avg	UNK-Avg	17.418	0.544	0.544	Avg of preceding 3 Samples
205	2D Day 14	UNK-Rep		0.605		
206	2D Day 14-2	UNK-Rep		0.604		
207	2D Day 14-3	UNK-Rep		0.604		
208	2D Day 14-Avg	UNK-Avg	19.75	0.604	0.604	Avg of preceding 3 Samples
209	3C Day 14	UNK-Rep		0.19		
210	3C Day 14-2	UNK-Rep		0.19		
211	3C Day 14-3	UNK-Rep		0.19		
212	3C Day 14-Avg	UNK-Avg	3.687	0.19	0.19	Avg of preceding 3 Samples
213	3D Day 14	UNK-Rep		0.02		
214	3D Day 14-2	UNK-Rep		0.02		
215	3D Day 14-3	UNK-Rep		0.02		
216	3D Day 14-Avg	UNK-Avg	-2.897	0.02	0.02	Avg of preceding 3 Samples